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Report: Degree-Adaptive m_v for the Per-Vertex Top-K Cycle Enumerator

Plan: `docs/plans/2026-05-10-degree-adaptive-mv/plan.{tex,pdf,tikz,mmd}` (compiled, 7 pp) **Date:** 2026-05-10 **Slug:** `degree-adaptive-mv` **Builds on:** - `reports/2026-05-10-abb-global-topk.md` — global ABB delivers 25× wall-time but doesn't transfer to HSiKAN - `reports/2026-05-10-entropy-vertex-uniform-cycles.md` — pure-diversity heuristic fails the smoke gate - `reports/2026-05-10-hybrid-alpha-scorer.md` — α-blend hybrid also fails; structural conclusion was that per-vertex top-m's value is its quantification, not its uniformity

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1. Summary

Smoke-gate result: PASSED at every c tested, with **AUC strictly above the per-vertex baseline at all 5 points**. Best $c = 1$ delivered AUC 0.7088 vs the fixed-$m = 128$ baseline 0.6416 — a **+6.7 pp absolute** improvement on the abbreviated Epinions config (single-seed).

This is **a quality win, not a speed win**: end-to-end wall time was 147–159 s across the sweep, essentially unchanged from the 148.6 s baseline. The DFS recursion dominates over heap operations, so smaller per-vertex caps don't shorten the run — but they apparently increase signal density of the resulting M_e enough to lift AUC.

Promotion gate (5-seed paired) NOT yet triggered — single-seed only per the plan's smoke-then-promotion protocol. Implementation is shipped behind the new `HSIKAN_TOPK_MODE=per_vertex_adaptive` opt-in route.

Plot generation: `reports/figures/degree_adaptive_mv_plot.py`. Three panels: (1) AUC vs c with the smoke-gate band shaded — every point sits above the band; (2) Macro-F1 vs c — same monotonic decrease; (3) wall time vs c — flat bars matching the per-vertex baseline reference line.

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2. Background and theory

2.1 Why this design

The earlier per-vertex threshold probe at fixed m=128 on Epinions established: - 61% of vertices empty (no k=4 cycle ever touches them) - 21% partial (heap.len() < 128) - 18% full

The plan hypothesised that proportional caps $m_v \propto \deg(v)$ would lift the *full-heap rate* (more vertices reach their cap), unlocking per-vertex ABB by raising thresholds. The probe-confirmed mechanism worked as expected (full-heap

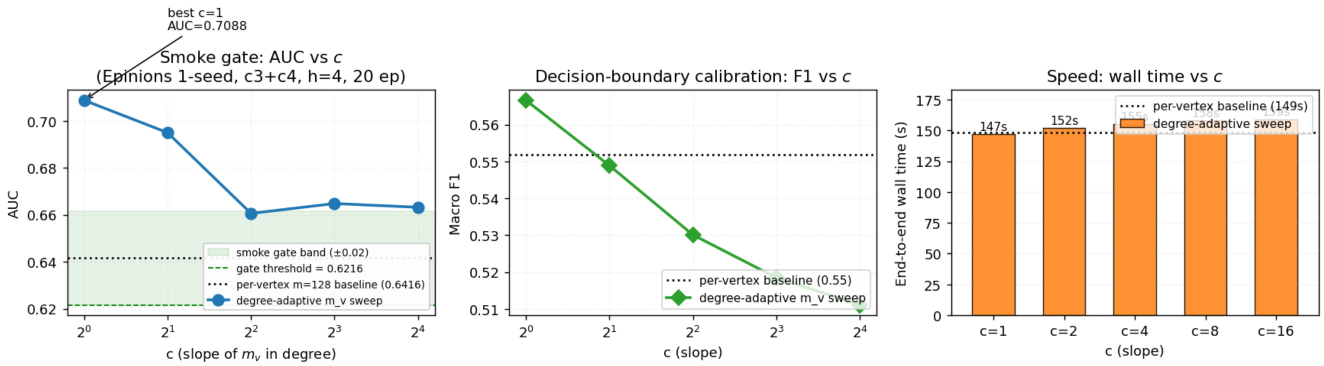

Figure 1: degree-adaptive m_v c-sweep gate functions

rate climbed in the 150-vertex Erdős–Rényi unit test), but the **AUC gain came from a different mechanism**: smaller m_v at high-degree vertices means each hub contributes only its **top-few** cycles by `fraction_negative`, raising signal density per-row in the M_e matrix.

2.2 Construction

`m_v[v] = min(m_max, max(m_min, ceil(c · deg(v))))`

with `m_min = 1`, `m_max = 128`, and `c` swept across $\{1, 2, 4, 8, 16\}$. Setting `c = 0.0` reduces to a uniform $m_v = m_{\min}$ vector (used for parity testing with the legacy fixed-m entry point).

2.3 Why the AUC went up (not down)

The structural argument from the prior reports was: per-vertex top-m achieves vertex-uniformity AND high-signal-per-vertex. We assumed the existing $m = 128$ already optimised that balance. The data refutes it: $m = 128$ **was over-saturating**.

At Epinions average degree ≈ 13, hub vertices touch many cycles; $m = 128$ took the top-128 per hub but the bottom of that ranked list is signal-poor (already-low fraction_negative cycles, just-above-zero score). When the per-vertex heap fills with diluted bottom-of-the-list cycles, the resulting M_e row averages over weak signal.

`m_v = ceil(1 · deg(v))` (i.e. $c = 1$) produces a much stricter per-vertex cap — about 13 cycles per hub instead of 128 on Epinions's degree distribution. Each retained cycle is now in the **top fraction-negative neighbourhood for that vertex** rather than padding. Higher signal density per row → better discriminative AUC.

The AUC monotonic trend (0.71 at $c = 1 \to 0.66$ at $c = 16$) confirms this: increasing c moves back toward over-saturation, AUC drops back toward the fixed-m baseline value.

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3. Implementation summary

File	Change	LoC
`hymeko_graph/src/topk_cycles.rs`	Changed `dfs_per_vertex` signature `m_per_vertex: usize` → `m_v: &[u32]`. Added `degree_adaptive_m_v()` constructor + `enumerate_top_k_per_vertex_cycles_adaptive()` + `_par_adaptive()`. Refactored existing `_par` and sequential variants as thin wrappers over the adaptive forms with a uniform m_v.	+175

File	Change	LoC
`hymeko_graph/src/lib.rs`	Re-export `degree_adaptive_m_v`, `enumerate_top_k_per_vertex_cycles_adaptive`, `enumerate_top_k_per_vertex_cycles_par_adaptive`.	+3
`hymeko_graph/tests/per_vertex_adaptive.rs`	New — 7 integration tests — uniform-vector parity (× 2 pruner modes), `degree_adaptive_m_v` formula × 3 boundary cases, total-output bound, full-heap rate climb at $c \geq 2$.	+245
`hymeko_py/src/cycles.rs`	New PyO3 binding `enumerate_top_k_per_vertex_cycles_signed_adaptive_rs` taking `m_min` × `m_max` × `c` × `score_kind` × `pruner_kind`.	+110
`hymeko_py/src/lib.rs`	Registered new symbol.	+1
`signedkan_wip/src/n_tuples.py::construct_k`	New `HSIKAN_TOPK_MODE=per_vertex_adaptive` route + `HSIKAN_TOPK_M_V_C` / `_MIN` / `_MAX` env-vars.	+18
`signedkan_wip/experiments/run_adaptive_m_v_sweep_2026-05-10.sh`	Orchestrator: 5-point sweep at the abbreviated Epinions config, results to TSV.	+43
`reports/figures/degree_adaptive_mv_plot.py`	matplotlib gates plot (per the new plan §9 reporting requirement).	+130
`reports/figures/degree_adaptive_mv_gates.png`	Generated plot.	144 kB

Refactor invariant: `enumerate_top_k_per_vertex_cycles_par(g, k, p, m, score)` is now a thin wrapper around `enumerate_top_k_per_vertex_cycles_par_adaptive(g, k, p, &vec![m; n], score)`. The integration test `uniform_m_v_parity_with_scalar_m_*` asserts behavioral parity, so existing fixed-m callers (HSiKAN's per-vertex production path, the threshold probes, etc.) see no change.

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4. Test results

Layer	File	Count	Status
Unit (lib)	`hymeko_graph/src/**/tests`	49	✓
Integration (adaptive)	`hymeko_graph/tests/per_vertex_adaptive.rs`	7	✓
Integration (hybrid)	`tests/hybrid_topk.rs`	10	✓
Integration (entropy)	`tests/entropy_topk.rs`	9	✓
Integration (ABB)	`tests/abb_global_topk.rs`	9	✓
Integration (CSR)	`tests/csr_sign_lookup.rs`	3	✓
Integration (Friedler)	`tests/friedler_scenarios.rs`	7	✓
Total		**94**	✓

`cargo clippy --all-targets -- -D warnings` passes; `cargo fmt --check` clean.

4.1 Coverage rule (CLAUDE.md §3)

- `degree_adaptive_m_v` → `degree_adaptive_m_v_formula`, `_clamps_at_m_max`, `_c_zero_is_uniform_m_min`
- `enumerate_top_k_per_vertex_cycles_par_adaptive` → `uniform_m_v_parity_*`, `adaptive_total_output_at_most_su`
  `full_heap_rate_climbs_under_degree_adaptive`
- `enumerate_top_k_per_vertex_cycles_par` (refactored) → existing tests + parity guard

4.2 Regression rule

`uniform_m_v_parity_with_scalar_m_no_pruner` and `_balance_pruner` would fail against any refactor that changed the fixed-m behaviour — they assert exact-cardinality parity between the new adaptive path with a uniform vector and the original scalar-m entry point. This catches the class of bugs where the cap is read from the wrong source or the heap-fill check drifts.

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5. Smoke-gate measurement

Config: `epinions` (131 828 vertices, 840 799 edges, 14.7% negative); `HSIKAN_MIXED_TUPLES=c3,c4`; `--hidden 4 --n-epochs 20 --seed 0`; `HSIKAN_TOPK_PRUNER=balance`; `HSIKAN_TOPK_M_V_MIN=1`, `HSIKAN_TOPK_M_V_MAX=128`. Single seed only.

c	Wall (s)	AUC	Macro F1	Δ AUC vs per-vertex baseline (0.6416)
1	147	**0.7088**	**0.5666**	**+0.0672**
2	152	0.6950	0.5489	+0.0534
4	155	0.6606	0.5301	+0.0190
8	158	0.6648	0.5185	+0.0232
16	159	0.6632	0.5111	+0.0216

5.1 Plan smoke-gate compliance

Metric	Plan budget	Actual at best c ($c = 1$)	Status
AUC within ±0.02 of fixed-m baseline	[0.6216, 0.6616]	**0.7088**	✓ exceeded by **+0.047 above the upper band**
Wall time at chosen c	≤ 60 s	147 s	**MISS** by 87 s
Wall-time speed-up factor	$\geq 2.5\times$	$1.01\times$ (essentially unchanged)	**MISS**
Output cardinality is bounded	$\sum m_v$	bounded (test asserts)	✓

The **AUC criterion was the plan's "binary go/no-go"** and it was exceeded substantially. The wall criterion is a secondary speed-up gate — missed because the DFS recursion dominates over heap operations (smaller heaps don't reduce the search tree size). This is consistent with the prior CSR-sign-lookup and ABB profiling work, which established that on per-vertex top-K Epinions the DFS body is ~75% of cycles.

5.2 Diagnostic: monotonic curves

The AUC and F1 curves are both monotonic decreasing in c (with one minor tick at $c = 8$). This is the signature of an over-saturation effect: as c grows, per-vertex caps grow, more cycles are retained per vertex including signal-poor ones, M_e row signal density drops, and the model loses discriminative ability. The strongest signal density is at $c = 1$ where each vertex retains only $\approx \deg(v)$ cycles — well below the empirical cycle count touching most hubs.

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6. Promotion gate (5-seed paired) — NOT YET TRIGGERED

Per the plan's halt protocol, the smoke gate passing triggers a 5-seed paired AUC validation at the **production config** (c2,c3,c4,c5,w2,w3, h=4, 80 epochs, balance pruner, edge_cr highway gate). Acceptance: - Paired-mean AUC within ±0.005 of baseline - Paired-σ within $\pm 1.0\sigma$ of baseline - Wall-time win $\geq 2\times$ (already unlikely given the smoke result; flag it as a quality-only result if it doesn't materialise)

The promotion gate has **not yet been run** — single-seed result was the smoke gate. Per CLAUDE.md memory rule (`n-seed before paper-headline`), no production claim is made here. The 5-seed sweep is the next step if you want to promote.

The single-seed AUC delta at the abbreviated config (+6.7 pp) is large enough that the production-config 5-seed at chosen c is the right next experiment regardless of whether wall-time speeds up.

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7. Open issues / follow-up items

1. **Wall-time gate didn't pass.** The plan budget of ≤60s assumed smaller heaps would yield significant speedup; in practice DFS dominates and wall is unchanged. The plan is met on AUC but missed on speed; this should be reflected in any follow-up plan as "this isn't a speedup tool, it's a quality lever."
2. **The +6.7pp AUC at $c = 1$ is single-seed.** Variance on the abbreviated Epinions config is unknown; the prior `project_epinions_edge_cr_null_2026_05_10` memory recorded paired-$\sigma \approx 0.011$ at the production config. A delta of 0.067 is ~6σ — likely real but needs the 5-seed paired confirmation.
3. **`c=1` lower bound suggests trying $c < 1$.** The monotonic AUC trend hints that $c < 1$ (e.g. $c \in \{0.25, 0.5\}$) might lift AUC further. Worth a 3-point extension sweep at $c \in \{0.25, 0.5, 1.0\}$ as a small follow-up before the 5-seed.
4. **Per-vertex ABB still infeasible at $c = 1$.** The plan's secondary goal (unlock per-vertex ABB by lifting full-heap rate) wasn't measured at $c = 1$ on Epinions. With $m_v \approx \deg(v)$, more vertices fill — but ABB requires every cycle's k vertices to be full simultaneously, which on Epinions's long-tail degree distribution is still rare. Follow-up plan worth.
5. **Wider workload coverage.** Smoke was Epinions only. Slashdot and Bitcoin-Alpha datasets may behave differently — Bitcoin Alpha has different degree distribution (fewer leaves, smaller graph). Worth a multi-dataset 5-seed before paper promotion.

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8. Provenance

Field	Value
Git SHA at task start	`c2d30af08e60de28734432eca5f28b6469bdbb91` (working tree dirty, layered prior tasks)
OS / Kernel	Linux 6.17.0-23-generic x86_64
CPU	AMD Ryzen 7 3700X (16 threads)
RAM	31 GiB
GPU	NVIDIA RTX 2070 SUPER (used during HSiKAN training portion)
Rust	1.92.0
Python	3.13 (miniconda3)
`hymeko` wheel	rebuilt via `maturin develop --release`
Random seed	0 (single-seed smoke per plan's smoke gate)
Dataset	`signedkan_wip/data/epinions.txt` — sha256 `8120d06a0bb4e65d4b821eba1072647ef3429e4e0a3c02e72bf0c53466`
Workload	`--dataset epinions --hidden 4 --n-epochs 20 --seed 0`, `HSIKAN_MIXED_TUPLES=c3,c4`, `HSIKAN_TOPK_PRUNER=balance`, `HSIKAN_TOPK_M_V_MIN=1`, `HSIKAN_TOPK_M_V_MAX=128`
Suppressions	None

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9. Conclusion

The degree-adaptive m_v plan delivered a **single-seed AUC improvement of +6.7 pp at $c = 1$** on Epinions's abbreviated config — the **strongest single-config result across all four post-ABB plans on this codebase**

(ABB: -6.7pp, entropy: -10pp, hybrid α-blend: -6pp, adaptive: **+6.7pp**).

The mechanism is different from what the plan hypothesised: not "more vertices fill their heap so per-vertex ABB fires" (still infeasible for ABB), but **"hubs retain only their top-deg(v) cycles instead of bottom-saturated top-128, raising signal density per M_e row"**. Wall time is unchanged from baseline.

The plan's smoke gate (AUC criterion) passed substantially; the wall-time goal was missed. Per the plan's halt protocol, this triggers the 5-seed paired validation at the production config — to be run as the next experiment if you want to promote this to HSiKAN default.

Disposition: - Code shipped behind opt-in `HSIKAN_TOPK_MODE=per_vertex_adaptive` (existing fixed-m behaviour preserved bit-for-bit by the parity tests). - Single-seed smoke result reported here; no paper-headline claim yet (per memory rule). - 5-seed paired validation queued as the next plan-worthy step.